- **CUSTOMER CHURN PREDICTION PROJECT**

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OVERVIEW

- The company is experiencing customer churn
- Losing customers leads to revenue loss
- We analyzed customer data to identify:
  - Customers likely to churn
  - Key factors influencing churn
- The goal is to use data to support better business decisions

Project Objective

- Develop a predictive model for churn
- Support business decision-making
- Improve customer retention strategies

BUSINESS UNDERSTANDING

- **PROBLEM**
- High customer churn
- Loss of revenue
- Difficult to identify at-risk customers
- Decisions currently not data-driven
- **QUESTION**
- 👉 Which customers are most likely to churn?
  👉 How can we predict and reduce churn?

PROJECT GOALS

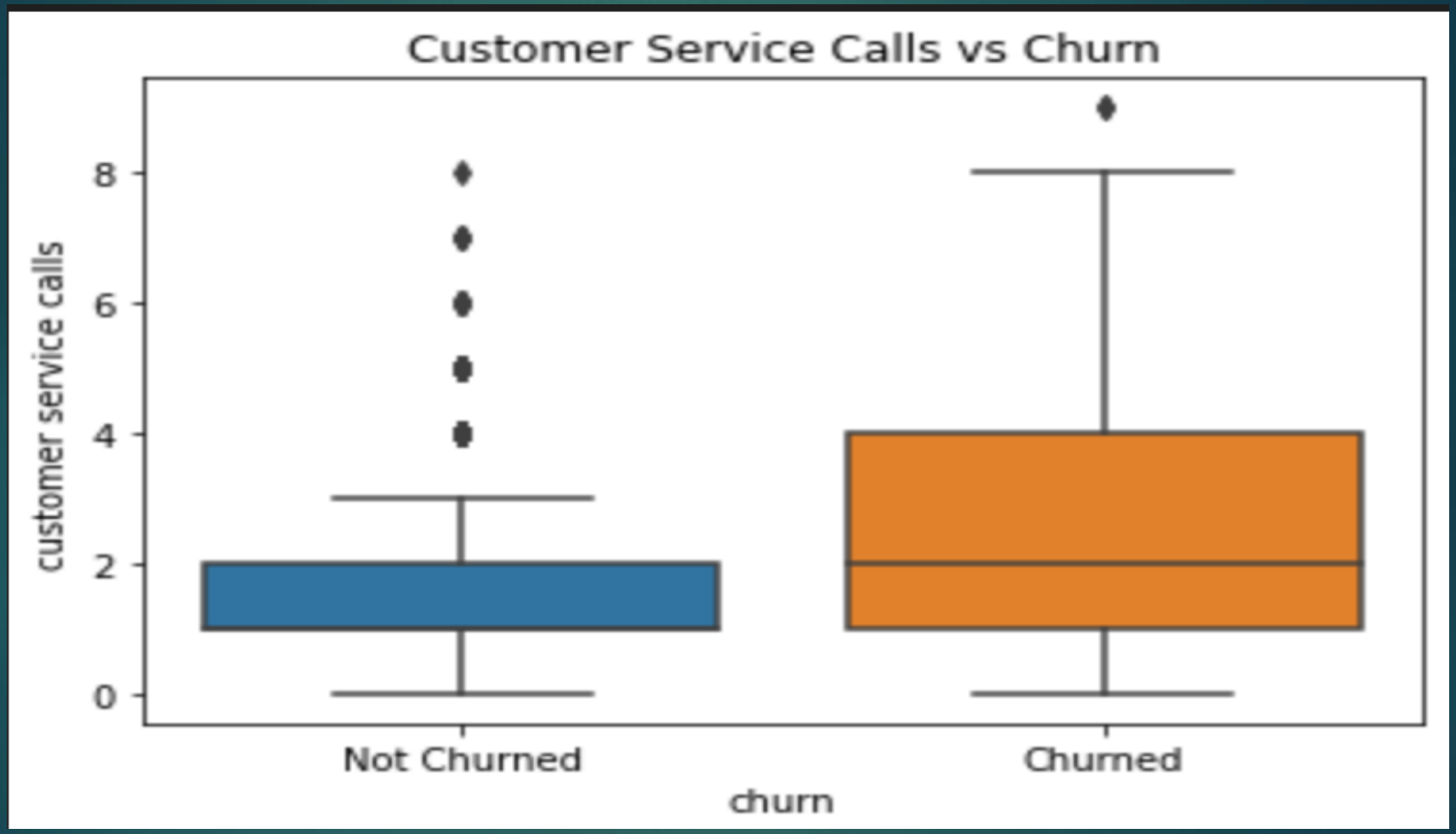
- **FIRST**
- Predict customer churn using machine learning

- **SECOND**
- Provide insights to improve customer retention

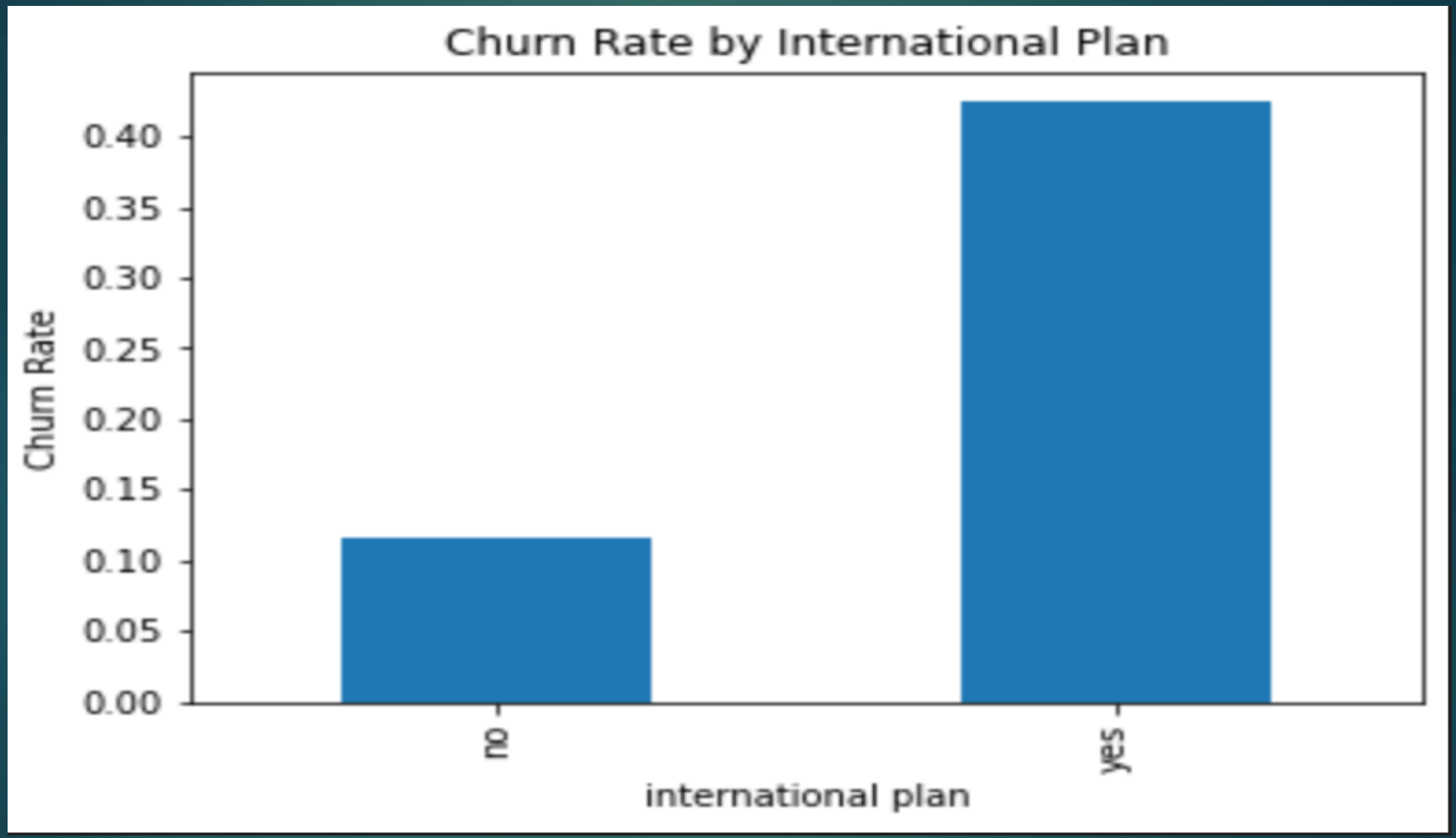
DATA PREPARATION

- **STEP 1 — Data Cleaning**
- Removed duplicates
- Handled missing values
- Fixed inconsistent data
- **STEP 2 — Data Transformation**
- Encoding categorical variables
- Scaling numerical features
- **STEP 3 — Feature Preparation**
- Selected relevant variables
- Prepared data for modeling

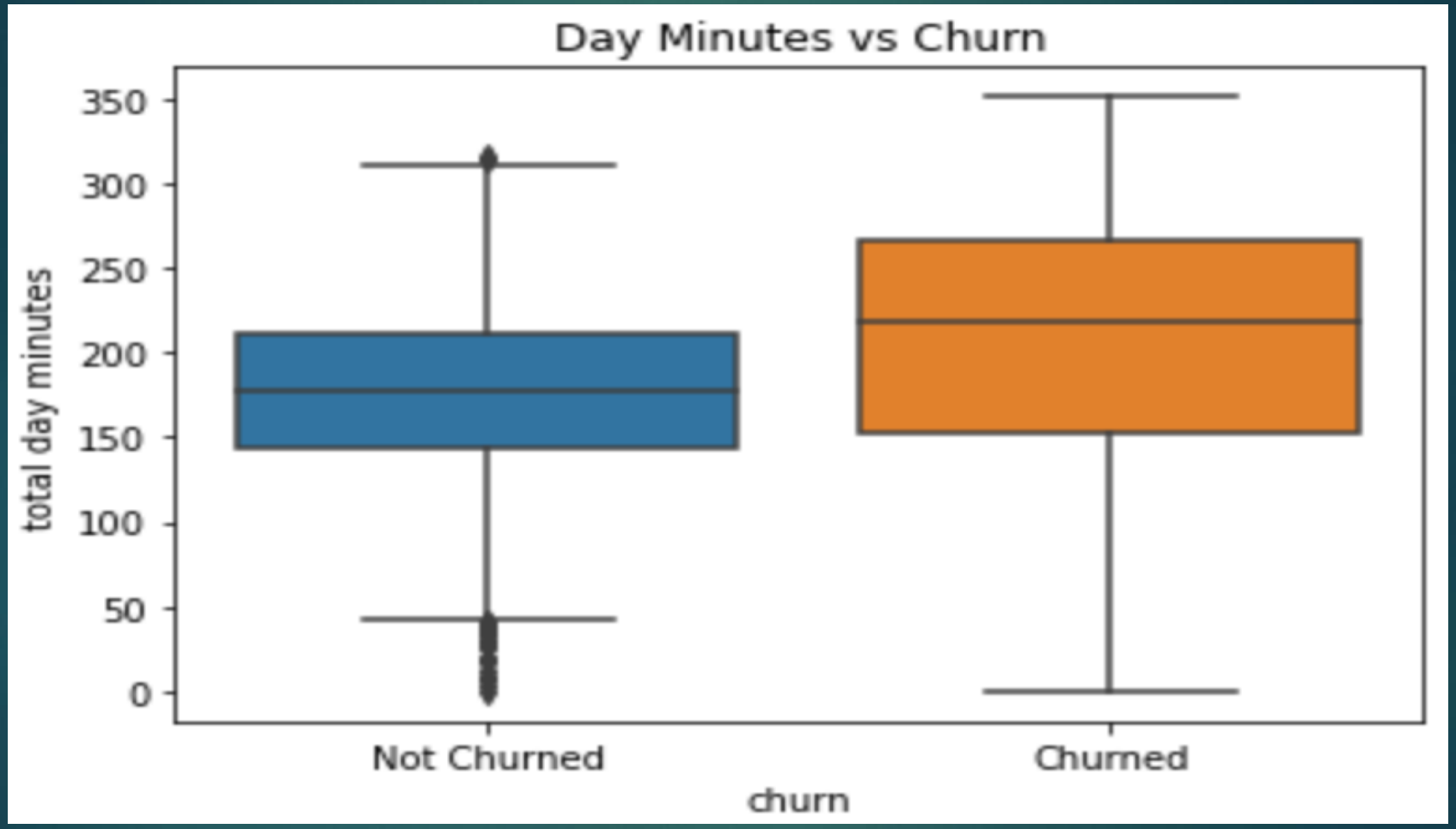
EDA1


Customer Service Calls vs Churn
customer service calls
0
2
4
6
8
Not Churned
Churned
churn

EDA2


Churn Rate by International Plan
Churn Rate
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
no
yes
international plan

EDA3


Day Minutes vs Churn
total day minutes
350
300
250
200
150
100
50
0
Not Churned
Churned
churn

Results:EDAs

- **EDA1**
- Customers with frequent service calls are more likely to churn, indicating poor customer experience.
- **EDA2**
- Having an international plan is associated with a higher likelihood of churn.
- **EDA3**
- Higher usage customers tend to churn more, suggesting cost or service-related dissatisfaction.

Model Comparison

- **Logistic Regression (Original)**
- High accuracy
- Very low recall for churn
- **Logistic Regression (SMOTE)**
- Improved recall
- Slight drop in accuracy
- **Decision Tree (Without SMOTE)**
- High accuracy (~94%)
- Strong AUC (~0.91)
- Balanced performance
- **Decision Tree (With SMOTE)**
- Higher recall
- Lower accuracy (~80%)
- More false positives

Model selection depends on balancing accuracy and recall

Final Model Selection

- **Decision Tree (Without SMOTE)**
- Best overall performance
- High accuracy
- Strong AUC
- Good balance between detecting churn and avoiding errors

Chosen because it provides the best balance between performance and interpretability

Business Impact

- Identify customers at risk of churn
- Enable proactive retention strategies
- Improve revenue and customer satisfaction

Recommendations

- **Focus on At-Risk Customers**
  - Target customers predicted as churn
- **Improve Retention Strategies**
  - Offer incentives and better support
- **Monitor Model Performance**
  - Continuously update with new data
- **Balance Business Trade-offs**
  - Choose between accuracy and recall
- **Analyze Key Features**
  - Identify main drivers of churn
- **Explore Future Models**
  - Test advanced models for improvement

CONCLUSION

- Our analysis shows that customer churn can be effectively predicted using machine learning.
- The results show that:
- Customer behavior patterns strongly influence churn
- Model performance depends on handling class imbalance
- Decision Tree provided the best balance of performance
- 👉 Using this model, the company can make better, data-driven decisions

This solution enables proactive decision-making instead of reactive actions